

Trajectory Classification with LSTM for Robotic Peg-in-Hole

Group 1 Project

UE5 Apprentissage

Master's Machine Learning
Project – 2025–2026





Project Objectives

- Build a trajectory-level classifier for a 7-DoF peg-in-hole task with **3 classes**.



Project Objectives

- ▶ Build a trajectory-level classifier for a 7-DoF peg-in-hole task with **3 classes**.
- ▶ Use a sequence model aligned with course material (**RNN/LSTM** module).



Project Objectives

- ▶ Build a trajectory-level classifier for a 7-DoF peg-in-hole task with **3 classes**.
- ▶ Use a sequence model aligned with course material (**RNN/LSTM** module).
- ▶ Deliver two outcomes required by the assignment:



Project Objectives

- ▶ Build a trajectory-level classifier for a 7-DoF peg-in-hole task with **3 classes**.
- ▶ Use a sequence model aligned with course material (**RNN/LSTM** module).
- ▶ Deliver two outcomes required by the assignment:
 - ▶ strong performance on the internal test split,



Project Objectives

- ▶ Build a trajectory-level classifier for a 7-DoF peg-in-hole task with **3 classes**.
- ▶ Use a sequence model aligned with course material (**RNN/LSTM** module).
- ▶ Deliver two outcomes required by the assignment:
 - ▶ strong performance on the internal test split,
 - ▶ explicit analysis of **feature influence** on performance.



Problem Framing

- Input: one full trajectory tensor $\in \mathbb{R}^{150 \times 20}$.



Problem Framing

- ▶ Input: one full trajectory tensor $\in \mathbb{R}^{150 \times 20}$.
- ▶ Output: one class label $\in \{0, 1, 2\}$ for the complete sequence.



Problem Framing

- ▶ Input: one full trajectory tensor $\in \mathbb{R}^{150 \times 20}$.
- ▶ Output: one class label $\in \{0, 1, 2\}$ for the complete sequence.
- ▶ Why sequence learning: class information is encoded in **temporal dynamics**, not only static snapshots.



Problem Framing

- ▶ Input: one full trajectory tensor $\in \mathbb{R}^{150 \times 20}$.
- ▶ Output: one class label $\in \{0, 1, 2\}$ for the complete sequence.
- ▶ Why sequence learning: class information is encoded in **temporal dynamics**, not only static snapshots.
- ▶ Evaluation context: internal split (train/val/test) in notebooks, plus compatibility with teacher-held hidden test format.



Roadmap

1. Data extraction and preparation



Roadmap

1. Data extraction and preparation
2. Model



Roadmap

1. Data extraction and preparation
2. Model
3. Experimental plan (hyperparameters)



Roadmap

1. Data extraction and preparation
2. Model
3. Experimental plan (hyperparameters)
4. Challenges encountered



Roadmap

1. Data extraction and preparation
2. Model
3. Experimental plan (hyperparameters)
4. Challenges encountered
5. Results and analysis



Roadmap

1. Data extraction and preparation
2. Model
3. Experimental plan (hyperparameters)
4. Challenges encountered
5. Results and analysis
6. Feature consequences study



Roadmap

1. Data extraction and preparation
2. Model
3. Experimental plan (hyperparameters)
4. Challenges encountered
5. Results and analysis
6. Feature consequences study
7. Conclusion



Dataset and Feature Blocks

- Files: `class1_trajectories.npy`, `class2_trajectories.npy`, `class3_trajectories.npy`

Block	Meaning	Size
peg_pose	peg position + quaternion	7
peg_vel	linear + angular velocity	6
hole_pos	hole position + quaternion	7
Total		20



Dataset and Feature Blocks

- ▶ Files: `class1_trajectories.npy`, `class2_trajectories.npy`, `class3_trajectories.npy`
- ▶ Balanced data: 256 trajectories/class, total 768 trajectories.

Block	Meaning	Size
peg_pose	peg position + quaternion	7
peg_vel	linear + angular velocity	6
hole_pos	hole position + quaternion	7
Total		20



Dataset and Feature Blocks

- ▶ Files: `class1_trajectories.npy`, `class2_trajectories.npy`, `class3_trajectories.npy`
- ▶ Balanced data: 256 trajectories/class, total 768 trajectories.
- ▶ Shape per sample: 150 timesteps, 20 features.

Block	Meaning	Size
peg_pose	peg position + quaternion	7
peg_vel	linear + angular velocity	6
hole_pos	hole position + quaternion	7
Total		20



Dataset and Feature Blocks

- ▶ Files: `class1_trajectories.npy`, `class2_trajectories.npy`, `class3_trajectories.npy`
- ▶ Balanced data: 256 trajectories/class, total 768 trajectories.
- ▶ Shape per sample: 150 timesteps, 20 features.
- ▶ Labels: `class1` $\rightarrow$ 0, `class2` $\rightarrow$ 1, `class3` $\rightarrow$ 2.

Block	Meaning	Size
<code>peg_pose</code>	peg position + quaternion	7
<code>peg_vel</code>	linear + angular velocity	6
<code>hole_pos</code>	hole position + quaternion	7
Total		20



Preprocessing Pipeline

- Reproducibility: fixed random seed (SEED=42) for split and initialization.

$$\mathbf{x}_{norm} = \frac{\mathbf{x} - \mu_{train}}{\sigma_{train} + \epsilon}$$



Preprocessing Pipeline

- ▶ Reproducibility: fixed random seed (SEED=42) for split and initialization.
- ▶ Split: **80/10/10** $\Rightarrow$ train 614, val 76, test 78.

$$\mathbf{x}_{norm} = \frac{\mathbf{x} - \mu_{train}}{\sigma_{train} + \epsilon}$$



Preprocessing Pipeline

- ▶ Reproducibility: fixed random seed (SEED=42) for split and initialization.
- ▶ Split: **80/10/10** $\Rightarrow$ train 614, val 76, test 78.
- ▶ Normalization: z-score computed on train only, then reused on val/test.

$$\mathbf{x}_{norm} = \frac{\mathbf{x} - \mu_{train}}{\sigma_{train} + \epsilon}$$



Preprocessing Pipeline

- ▶ Reproducibility: fixed random seed (SEED=42) for split and initialization.
- ▶ Split: **80/10/10** $\Rightarrow$ train 614, val 76, test 78.
- ▶ Normalization: z-score computed on train only, then reused on val/test.
- ▶ Numerical stability: $\sigma = 0$ replaced by small ϵ .

$$\mathbf{x}_{norm} = \frac{\mathbf{x} - \mu_{train}}{\sigma_{train} + \epsilon}$$



Preprocessing Pipeline

- ▶ Reproducibility: fixed random seed (SEED=42) for split and initialization.
- ▶ Split: **80/10/10** $\Rightarrow$ train 614, val 76, test 78.
- ▶ Normalization: z-score computed on train only, then reused on val/test.
- ▶ Numerical stability: $\sigma = 0$ replaced by small ϵ .

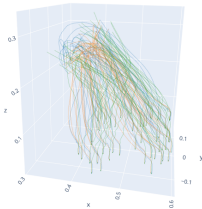
$$\mathbf{x}_{norm} = \frac{\mathbf{x} - \mu_{train}}{\sigma_{train} + \epsilon}$$

- ▶ This prevents leakage and improves optimization conditioning.

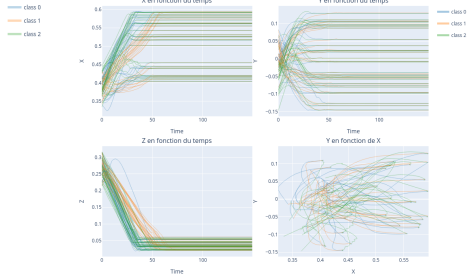


Trajectory Evidence Before Training

Trajectoires (position du peg) par classe



Trajectoires par composante



3D trajectories (x, y, z) by class.

Projections: $x(t), y(t), z(t), y(x)$.

Class-dependent temporal patterns are visible before training, which motivates a temporal classifier.



Why LSTM (Course Connection)

- Course reminder (3_RNN_Transformers_2025.pdf): vanilla RNNs are sensitive to vanishing/exploding gradients through BPTT.

$$i_t, f_t, o_t = \sigma(\cdot), \quad \tilde{c}_t = \tanh(\cdot), \quad c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$$



Why LSTM (Course Connection)

- ▶ Course reminder (3_RNN_Transformers_2025.pdf): vanilla RNNs are sensitive to vanishing/exploding gradients through BPTT.
- ▶ LSTM uses gated memory to control information flow and preserve useful long-range signals.

$$i_t, f_t, o_t = \sigma(\cdot), \quad \tilde{c}_t = \tanh(\cdot), \quad c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$$



Why LSTM (Course Connection)

- ▶ Course reminder (3_RNN_Transformers_2025.pdf): vanilla RNNs are sensitive to vanishing/exploding gradients through BPTT.
- ▶ LSTM uses gated memory to control information flow and preserve useful long-range signals.

$$i_t, f_t, o_t = \sigma(\cdot), \quad \tilde{c}_t = \tanh(\cdot), \quad c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$$

- ▶ This architecture matches the project setting: long trajectories (150 steps) with dynamic class signatures.



Final Architecture and Training Setup

- ▶ Input tensor: $(\text{batch}, 150, 20)$.



Final Architecture and Training Setup

- ▶ Input tensor: (batch, 150, 20).
- ▶ Backbone: LSTM(num_layers=2, hidden_size=64, dropout=0.2).



Final Architecture and Training Setup

- ▶ Input tensor: (batch, 150, 20).
- ▶ Backbone: LSTM(num_layers=2, hidden_size=64, dropout=0.2).
- ▶ Classifier head: last hidden state $\rightarrow$ Linear(64,3).



Final Architecture and Training Setup

- ▶ Input tensor: (batch, 150, 20).
- ▶ Backbone: LSTM(num_layers=2, hidden_size=64, dropout=0.2).
- ▶ Classifier head: last hidden state $\rightarrow$ Linear(64, 3).
- ▶ Loss/optimizer: CrossEntropyLoss + Adam(lr=1e-3).



Final Architecture and Training Setup

- ▶ Input tensor: (batch, 150, 20).
- ▶ Backbone: LSTM(num_layers=2, hidden_size=64, dropout=0.2).
- ▶ Classifier head: last hidden state $\rightarrow$ Linear(64,3).
- ▶ Loss/optimizer: CrossEntropyLoss + Adam(lr=1e-3).
- ▶ Batch size: 32, training budget up to 40 epochs in main experiments.



Final Architecture and Training Setup

- ▶ Input tensor: (batch, 150, 20).
- ▶ Backbone: LSTM(num_layers=2, hidden_size=64, dropout=0.2).
- ▶ Classifier head: last hidden state $\rightarrow$ Linear(64, 3).
- ▶ Loss/optimizer: CrossEntropyLoss + Adam(lr=1e-3).
- ▶ Batch size: 32, training budget up to 40 epochs in main experiments.
- ▶ Stability control: gradient clipping $\|\nabla\|_{max} = 1.0$.



Controlled Experiment Plan

Run	Setting	Purpose
R1	hidden=100, z-score, 10 epochs	baseline behavior
R2	hidden=64, no normalization	sensitivity to scaling
R3	hidden=64 + dropout	regularization impact
R4	R3 + clipping	stability impact

- Fixed controls: seed, split protocol, optimizer family, class definition.



Controlled Experiment Plan

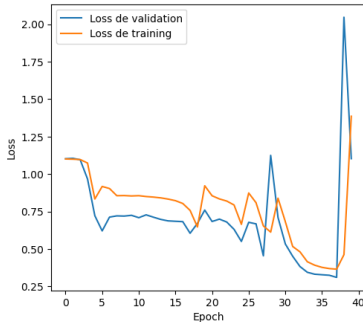
Run	Setting	Purpose
R1	hidden=100, z-score, 10 epochs	baseline behavior
R2	hidden=64, no normalization	sensitivity to scaling
R3	hidden=64 + dropout	regularization impact
R4	R3 + clipping	stability impact

- ▶ Fixed controls: seed, split protocol, optimizer family, class definition.
- ▶ Selection criterion: best generalization with stable learning curves.

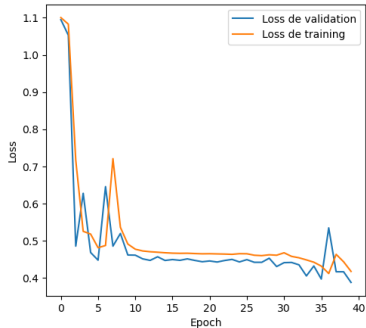


Practical Issues During Development

- **Reproducibility:** no fixed seed $\Rightarrow$ difficult run-to-run comparison.



No normalization: noisier and less predictable convergence.

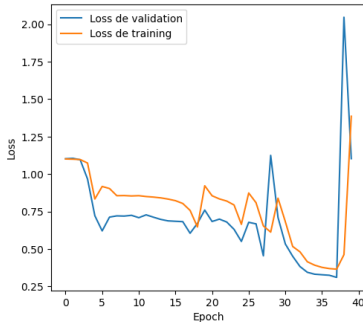


No dropout: weaker regularization and higher instability.

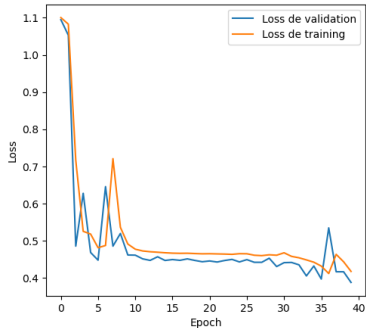


Practical Issues During Development

- **Reproducibility:** no fixed seed $\Rightarrow$ difficult run-to-run comparison.
- **Optimization sensitivity:** preprocessing and regularization strongly alter convergence quality.



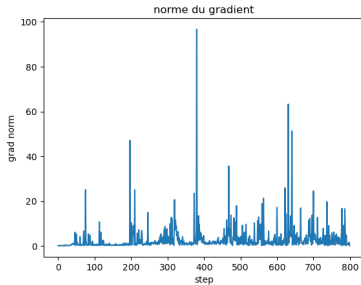
No normalization: noisier and less predictable convergence.



No dropout: weaker regularization and higher instability.



Exploding Gradients and Mitigation



Gradient norm spikes observed during training.

Observed issue

Large gradient spikes destabilize parameter updates.

Fix used

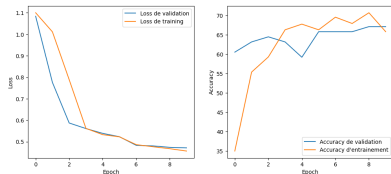
Gradient clipping in the training loop:
`clip_grad_norm(..., 1.0)`.

Effect

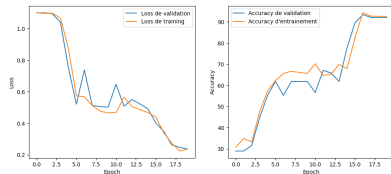
More stable updates and smoother final convergence.



Main Runs R1 and R2



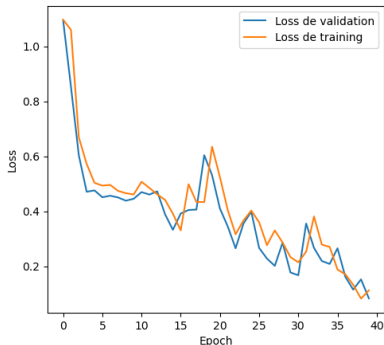
R1 baseline: limited/unstable behavior.



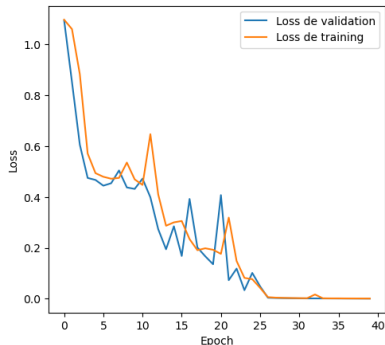
R2: strong gain, **92.31%** test accuracy (72/78).



Main Runs R3 and R4



R3 (+dropout): **98.72%** test accuracy
(77/78).



R4 (+dropout + clipping): **100%**
(78/78).



Quantitative Synthesis

Run	Test accuracy	Main effect
R1	baseline (unstable)	underfitting/instability
R2	92.31% (72/78)	better capacity choice
R3	98.72% (77/78)	dropout improves generalization
R4	100.00% (78/78)	clipping improves stability

- Most impactful additions in this pipeline: **dropout + clipping**.



Quantitative Synthesis

Run	Test accuracy	Main effect
R1	baseline (unstable)	underfitting/instability
R2	92.31% (72/78)	better capacity choice
R3	98.72% (77/78)	dropout improves generalization
R4	100.00% (78/78)	clipping improves stability

- ▶ Most impactful additions in this pipeline: **dropout + clipping**.
- ▶ Internal split performance is excellent; external hidden test remains the final robustness check.



Feature-Ablation Protocol (Feature Notebook)

Subset	Indices	Epoch budgets	Best test
Peg pose + orientation	0:7	40, 80, 150	100% (78/78)
Peg linear+angular velocity	7:13	40, 80, 150	100% (78/78)
Hole pose + orientation	13:20	40, 150	17.95% (14/78)

- Goal: isolate which observation blocks truly carry discriminative information.



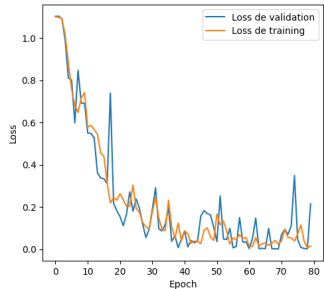
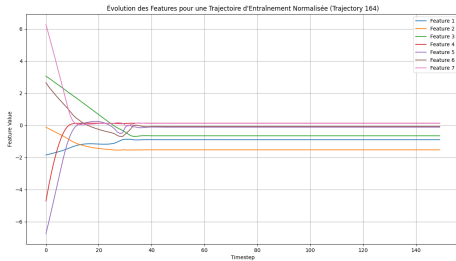
Feature-Ablation Protocol (Feature Notebook)

Subset	Indices	Epoch budgets	Best test
Peg pose + orientation	0:7	40, 80, 150	100% (78/78)
Peg linear+angular velocity	7:13	40, 80, 150	100% (78/78)
Hole pose + orientation	13:20	40, 150	17.95% (14/78)

- ▶ Goal: isolate which observation blocks truly carry discriminative information.
- ▶ Same LSTM family and evaluation protocol; only feature subset and epoch budget vary.



Subset A: Peg Pose/Orientation Only (0:7)



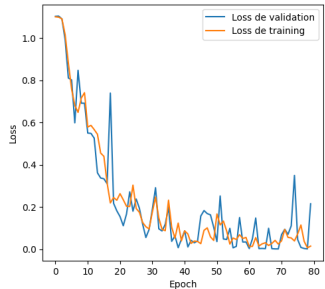
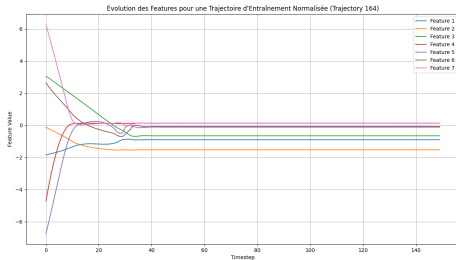
Evolution of the 7 peg pose/orientation features.

Example loss curve (80 epochs).

- Test performance: 98.72% (40 ep), 100% (80 ep), 100% (150 ep).



Subset A: Peg Pose/Orientation Only (0:7)



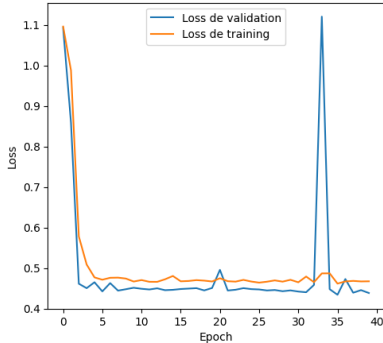
Evolution of the 7 peg pose/orientation features. Example loss curve (80 epochs).

- ▶ Test performance: 98.72% (40 ep), 100% (80 ep), 100% (150 ep).
- ▶ Interpretation: peg pose/orientation alone already contains most class-discriminative signal.

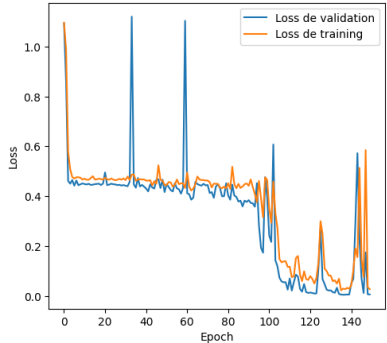
Feature Consequences Study



Subset B: Peg Velocities Only (7:13)



40 epochs: moderate, unstable plateau.

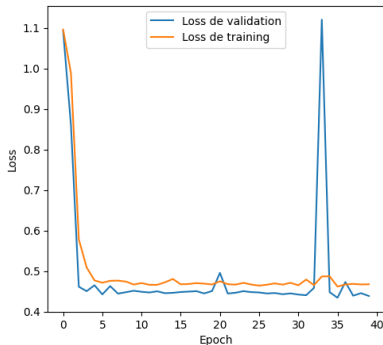


150 epochs: late transition to low loss.

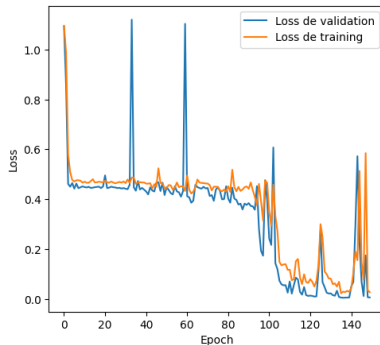
- Test performance: 64.10% (40 ep), 66.67% (80 ep), 100% (150 ep).



Subset B: Peg Velocities Only (7:13)



40 epochs: moderate, unstable plateau.

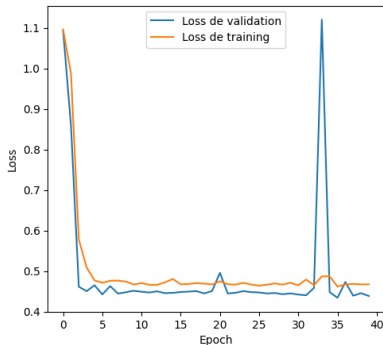


150 epochs: late transition to low loss.

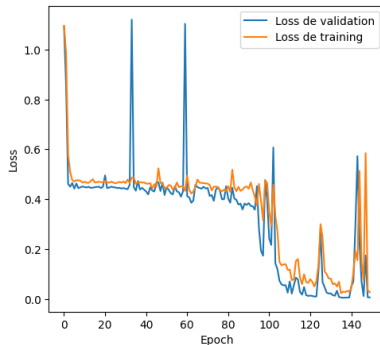
- ▶ Test performance: 64.10% (40 ep), 66.67% (80 ep), 100% (150 ep).
- ▶ Two regimes: long plateau, then sharp improvement around epochs ~ 95 –110.



Subset B: Peg Velocities Only (7:13)



40 epochs: moderate, unstable plateau.

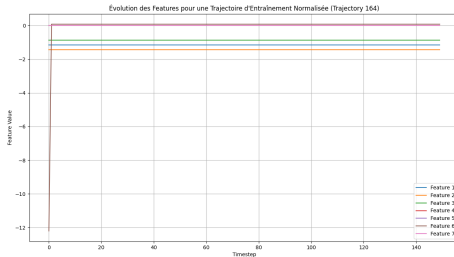


150 epochs: late transition to low loss.

- ▶ Test performance: 64.10% (40 ep), 66.67% (80 ep), 100% (150 ep).
- ▶ Two regimes: long plateau, then sharp improvement around epochs ~ 95 –110.
- ▶ Velocity is informative, but optimization is harder and slower.

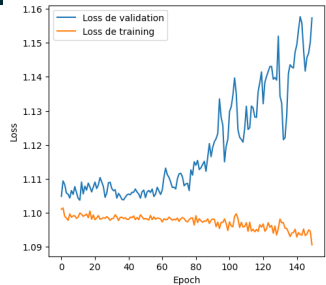


Subset C: Hole Pose/Orientation Only (13:20)



Hole-pose features remain quasi-constant over time.

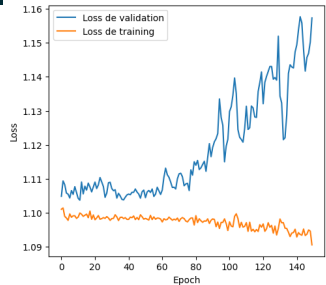
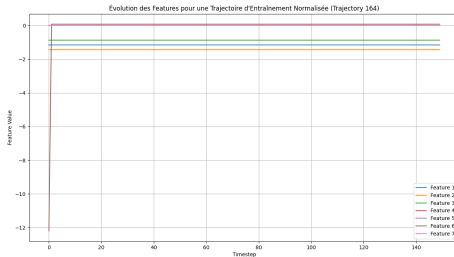
- Test performance drops: 32.05% (40 ep) → 17.95% (150 ep).



150 epochs: no meaningful learning.



Subset C: Hole Pose/Orientation Only (13:20)



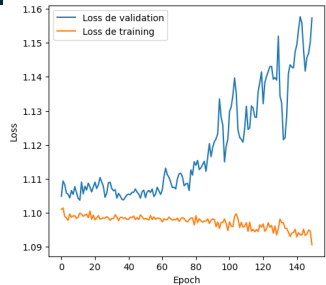
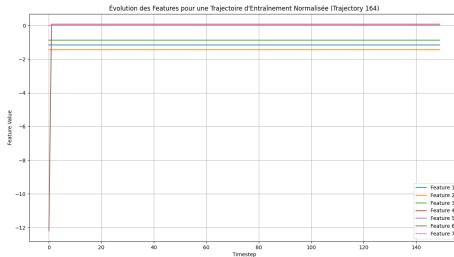
Hole-pose features remain quasi-constant over time.

150 epochs: no meaningful learning.

- ▶ Test performance drops: 32.05% (40 ep) → 17.95% (150 ep).
- ▶ Validation loss increases with training: more epochs worsen generalization.



Subset C: Hole Pose/Orientation Only (13:20)



Hole-pose features remain quasi-constant over time.

150 epochs: no meaningful learning.

- ▶ Test performance drops: 32.05% (40 ep) → 17.95% (150 ep).
- ▶ Validation loss increases with training: more epochs worsen generalization.
- ▶ Interpretation: this subset is non-discriminative for this task (near-constant signal).



Conclusion

- The final full-feature LSTM pipeline (2 layers, hidden 64, dropout, clipping) reaches top internal performance.



Conclusion

- ▶ The final full-feature LSTM pipeline (2 layers, hidden 64, dropout, clipping) reaches top internal performance.
- ▶ Methodical ablation clarifies causality:



Conclusion

- ▶ The final full-feature LSTM pipeline (2 layers, hidden 64, dropout, clipping) reaches top internal performance.
- ▶ Methodical ablation clarifies causality:
 - ▶ peg pose/orientation is highly informative,



Conclusion

- ▶ The final full-feature LSTM pipeline (2 layers, hidden 64, dropout, clipping) reaches top internal performance.
- ▶ Methodical ablation clarifies causality:
 - ▶ peg pose/orientation is highly informative,
 - ▶ peg velocity is informative but optimization-sensitive,



Conclusion

- ▶ The final full-feature LSTM pipeline (2 layers, hidden 64, dropout, clipping) reaches top internal performance.
- ▶ Methodical ablation clarifies causality:
 - ▶ peg pose/orientation is highly informative,
 - ▶ peg velocity is informative but optimization-sensitive,
 - ▶ hole pose/orientation alone is not discriminative here.



Conclusion

- ▶ The final full-feature LSTM pipeline (2 layers, hidden 64, dropout, clipping) reaches top internal performance.
- ▶ Methodical ablation clarifies causality:
 - ▶ peg pose/orientation is highly informative,
 - ▶ peg velocity is informative but optimization-sensitive,
 - ▶ hole pose/orientation alone is not discriminative here.
- ▶ Jury-level takeaway: good accuracy comes from both **signal selection** and **stable optimization**, not from architecture choice alone.



Conclusion

- ▶ The final full-feature LSTM pipeline (2 layers, hidden 64, dropout, clipping) reaches top internal performance.
- ▶ Methodical ablation clarifies causality:
 - ▶ peg pose/orientation is highly informative,
 - ▶ peg velocity is informative but optimization-sensitive,
 - ▶ hole pose/orientation alone is not discriminative here.
- ▶ Jury-level takeaway: good accuracy comes from both **signal selection** and **stable optimization**, not from architecture choice alone.
- ▶ Final validation step remains the teacher-held hidden test set.